

ATIF ABRAR BISWAS

[Email](#) | [LinkedIn](#) | [Google Scholar](#)

EDUCATIONAL QUALIFICATIONS

B.Sc. Industrial & Production Engineering
2026)

(March 2022-July

Rajshahi University of Engineering & Technology (RUET), Rajshahi
CGPA: 3.55 / 4.00

RESEARCH INTERESTS

Operation Research, Hesitant fuzzy set, Supply Chain, Quality Control, Operation Management, Product Design, Probability and Statistics, computational modeling.

TECHNICAL SKILLS

CAD: SolidWorks, Fusion 360, AutoCAD, Mastercam | **Programming:** C

Miscellaneous: MS Word, MS PowerPoint, MS Excel, LaTeX

PUBLICATIONS

- M. Ashaduzzaman, N. I. Nawar, S. A. Khan, **A. A. Biswas**, and M. M. Islam, “Strategic implementation of artificial intelligence and machine learning in the pharmaceutical sector: A dual hesitant fuzzy group decision making approach,” in *Proceedings of the 7th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, IEOM Society, 2024. <https://doi.org/10.46254/BA07.20240185>
- S. A. Khan, M. Ashaduzzaman, **A. A. Biswas**, N. I. Nawar, M. A. Haque, and S. Mahmud, “Integrating renewable energy and customer insights: A systematic approach to solar power bank design,” in *Proceedings of the 7th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, IEOM Society, 2024. <https://doi.org/10.46254/BA07.20240280>
- M. Ashaduzzaman, S. A. Khan, N. I. Nawar, **A. A. Biswas**, M. L. R. Lingkon, and M. S. Hossain, “Comparative analysis of forecasting methods and integration with inventory models for efficient demand management in the electronics sector,” in *SciEn Conference Series: Engineering*, vol. 3, pp. 627–632, 2025. <https://doi.org/10.38032/scse.2025.3.158>
- Optimizing Inventory by Minimizing Holding Costs and Stockout of Raw Materials of Denim Jacket Production Using LSTM Model: A Case Study, International Conference on Sustainable Supply Chain and Logistics Management (ICSSCLM-26) Seoul, South Korea, <https://academicresearchlibrary.com/abstract-detail/4888>

RESEARCH EXPERIENCES

Revolutionizing Healthcare System Through Artificial Intelligence Implementation: A Hesitant Probabilistic Decision-Making Approach | (Under Review):

- Developed SWARA to do the weighting of criteria and MSD to calculate the weight of experts in a hesitant probabilistic fuzzy system to assess AI implementation strategies. The combined expert-based review effectively placed the AI/ IoT training and diagnostic enrichment at the top for AI driven healthcare transformation.

Sizing a Sustainable Off-Grid Microgrid for River Island Electrification: The Case of Khidirpur Char |
(Ongoing Research)

- Optimized a grid-independent PV/Diesel/Battery hybrid renewable energy system for off-grid electrification (Khidirpur Char, Bangladesh) using HOMER Pro, NSGA-II, and MOPSO for tri-objective optimization (NPC, LCOE, CO₂); applied CRITIC–TOPSIS for weight-free compromise selection, achieving ~79% CO₂ reduction and 93% renewable fraction with zero unmet load.

Integrated Demand Forecasting and Inventory Management Model for Perishable Product
(Undergrad Thesis)

- Hybrid demand forecasting model improves demand forecasting
- Proposed constraint awareness inventory and integrated with demand forecasting to reduce product wastages and stock outs in retail shops.

Strategic Implementation of AI and ML in the Pharmaceutical Sector: A Dual Hesitant Fuzzy Group Decision Making Approach | *7th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management, 2024*

- Developed a dual hesitant multi criteria group decision making framework to evaluate AI and ML implementation strategies in the pharmaceutical sector and used the DHFHM operator to aggregate uncertain expert opinions and calculate strategy scores.

Integrating Renewable Energy and Customer Insights: A Systematic Approach to Solar Power Bank Design | *7th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management, 2024*

- Proposed a customer-driven solar power bank design using survey, kano model and QFD
- Selected optimal material using the Digital Logic Method
- Validated the design with CAD modeling and stress analysis which was structurally safe and feasible for manufacturing.

Comparative Analysis of Forecasting Methods and Integration with Inventory Models for Efficient Demand Management in the Electronics Sector | *8th International Conference on Mechanical, Industrial and Energy Engineering 2024*

- Compared three forecasting methods and found Linear Regression most accurate.
- Integrated forecasting with an inventory model (EOQ, safety stock, reorder level) to improve decision making.

PROJECTS

Mobile Phone Charger

Built a Mobile Phone Charger by using basic transformers and circuits for an academic lab course 'Electrical Engineering Lab'.

Solar Power Bank

Built a solar power bank that charges with both the help of solar and AC current as a part of the course 'Product Design Lab - II.'

Smart Rain Shield For Laundry: “Automated Protection with Arduino Uno”

Developed a fully automated laundry prototype that reduces manual labor and keeps clothes dry and protected in all weather

CERTIFICATIONS

- Participation in the research poster presentation competition at RUET—(Issued September 01, 2025)
- Participation in the event “Industrial visit to Sirajganj Power Plant operated by North-West Power Generation Company Limited (NWPGL), Sirajganj” organized by IEEE RUET—(Issued August 07, 2025)

AFFILIATION

- Society of Computer Aided Design RUET (SCADR)** (2024-2025)
Event Management Secretary & Cad Trainer
- Expertise in guiding design processes and executing well-coordinated, impactful events.
- RUET IPE CLUB, Rajshahi Bangladesh**
Executive Member, (2022- 2023)
- Organized events related to the club
 - Maintained connection with every member of the club
- Executive Member, Nobojagoron Foundation, a youth voluntary organization of Rajshahi University** (2025- present)
Executive Member
- Focuses on community development, providing educational support to underprivileged children in Rajshahi area

PROFESSIONAL EXPERIENCES

- Trainee – Square Toiletries Limited** (February 2025 – March 2025)
- Completed a structured trainee program at Square Toiletries Limited, gaining hands-on experience in FMCG operations and quality-focused work practices.
 - Assisted in daily production, quality checks, and process coordination while maintaining company standards and safety protocols.
 - Developed practical skills in teamwork, documentation, and problem-solving by supporting cross-functional departments in routine tasks.
- Industry Visit – North South Power Company Limited**
- Visited North South Power Company Limited to observe real-world power generation operations, gaining practical exposure to industrial processes and energy management.

REFERENCES

Dr. Md. Mohibul Islam Professor, Dept. of IPE, RUET Email Website	Dr. Shahed Mahmud Professor, Dept. of IPE, RUET Email Website
--	--

Dr. Md. Mohibul Islam

Professor, Dept. of IPE, RUET

[Email](#) | [Website](#)

Dr. Shahed Mahmud

Professor, Dept. of IPE, RUET [Email](#) |

[Website](#)